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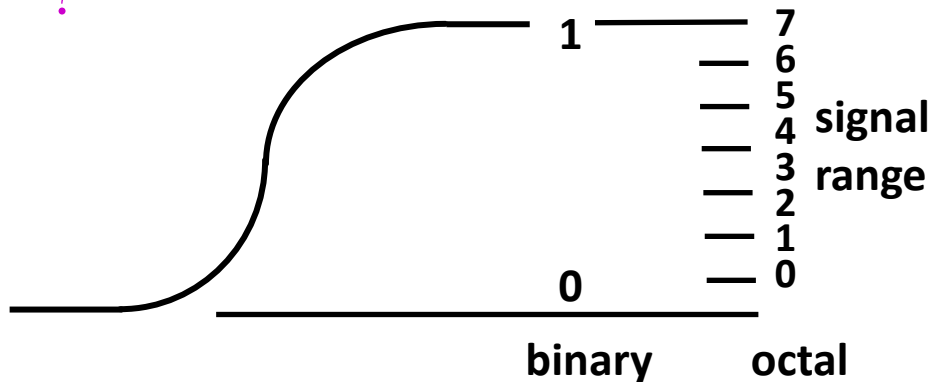
Introduction

❖ Introduction of Digital Electronics & Logic Circuits

• LOGIC GATES

• Digital Computers

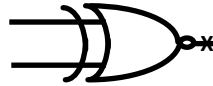
- - Imply that the computer deals with digital information, i.e., it deals with the information that is represented by binary digits
- - Why BINARY ? instead of Decimal or other number system ?



A single switch can be on or off, enabling the storage of 1 bit of information. Switches can be grouped together to store larger numbers. This is the key reason why binary is used in digital systems.

Introduction

COMBINATIONAL GATES

Name	Symbol	Function	Truth Table															
AND	A B 	$X = A \cdot B$ or $X = AB$	<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	X	0	0	0	0	1	0	1	0	0	1	1	1
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OR	A B 	$X = A + B$	<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	X	0	0	0	0	1	1	1	0	1	1	1	1
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I	A 	$X = A'$	<table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>0</td></tr></table>	A	X	0	1	1	0									
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Buffer	A 	$X = A$	<table><tr><th>A</th><th>X</th></tr><tr><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td></tr></table>	A	X	0	0	1	1									
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XOR Exclusive OR	A B 	$X = A \oplus B$ or $X = A'B + AB'$	<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td></tr></table>	A	B	X	0	0	0	0	1	1	1	0	1	1	1	0
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XNOR Exclusive NOR or Equivalence	A B 	$X = (A \oplus B)'$ or $X = A'B' + AB$	<table><tr><th>A</th><th>B</th><th>X</th></tr><tr><td>0</td><td>0</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr></table>	A	B	X	0	0	1	0	1	0	1	0	0	1	1	1
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Introduction of Number Systems

❖ Number Systems

- A number has two parts: Integer & Fractional parts.
- Positive radix or base, positional number systems
- A number with radix r is represented by a string of digits:

$A_{n-1}A_{n-2} \dots A_1A_0 . A_{-1}A_{-2} \dots A_{-m+1}A_{-m}$ in which $0 \leq A_i < r$ and $.$ is the radix point.

- The string of digits represents the power series:

$$\begin{array}{c} \text{(Number)}_r = \sum_{i=0}^{i=n-1} A_i \cdot r^i + \sum_{j=-m}^{j=-1} A_j \cdot r^j \\ \text{(Integer Portion)} \qquad \qquad \qquad + \qquad \qquad \text{(Fraction Portion)} \end{array}$$

Introduction of Number Systems

❖ Number Systems – Examples

	General	Decimal	Binary
Radix (Base)	r	10	2
Digits	$0 \Rightarrow r - 1$	$0 \Rightarrow 9$	$0 \Rightarrow 1$
Powers of Radix	0	r^0	1
	1	r^1	2
	2	r^2	4
	3	r^3	8
	4	r^4	16
	5	r^5	32
	-1	r^{-1}	0.5
	-2	r^{-2}	0.25
	-3	r^{-3}	0.125
	-4	r^{-4}	0.0625
	-5	r^{-5}	0.03125

Introduction of Number Systems

❖ Number Systems

- **The decimal number system** is said to be of base, or radix, 10 because it uses 10 digits and the coefficients are multiplied by powers of 10.
- **The binary system** is a different number system. The coefficients of the binary number system have only two possible values: 0 and 1.
- Each coefficient a_j is multiplied by a power of the radix, e.g., 2^j , and the results are added to obtain the decimal equivalent of the number.
- There are many different number systems. In general, a number expressed in a base- r system has coefficients multiplied by powers of r :

$$a_n \times r^n + a_{n-1} \times r^{n-1} + \dots + a_2 \times r^2 + a_1 \times r^1 + a_0 + a_{-1} \times r^{-1} + a_{-2} \times r^{-2} \\ + \dots + a_{-m} \times r^{-m}$$

Introduction of Number Systems

❖ Number Systems

- An example of a base-5 number is:

$$\begin{aligned}(4021.2)_5 &= 4 \times 5^3 + 0 \times 5^2 + 2 \times 5^1 + 1 \times 5^0 + 2 \times 5^{-1} \\ &= (511.4)_{10}\end{aligned}$$

- The coefficient values for base 5 can be only 0, 1, 2, 3, and 4. The octal number system is a base-8 system that has eight digits: 0, 1, 2, 3, 4, 5, 6, 7. An example of an octal number is 127.4. To determine its equivalent decimal value, we expand the number in a power series with a base of 8:

$$\begin{aligned}(127.4)_8 &= 1 \times 8^2 + 2 \times 8^1 + 7 \times 8^0 + 4 \times 8^{-1} \\ &= (87.5)_{10}\end{aligned}$$

Introduction of Number Systems

❖ Number Systems

- The letters of the alphabet are used to supplement the 10 decimal digits when the base of the number is greater than 10.
- The letters A, B, C, D, E, and F are used for the digits 10, 11, 12, 13, 14, and 15, respectively. An example of a hexadecimal number is:

$$\begin{aligned}(B65F)_{16} &= 11 \times 16^3 + 6 \times 16^2 + 5 \times 16^1 + 15 \times 16^0 \\ &= (46687)_{10}\end{aligned}$$

- The hexadecimal system is used commonly by designers to represent long strings of bits in the addresses, instructions, and data in digital systems. For example, B65F is used to represent 1011011001010000.
- When a bit is equal to 0, it does not contribute to the sum during the conversion. Therefore, the conversion from binary to decimal can be obtained by adding only the numbers with powers of two corresponding to the bits that are equal to 1. For example,

$$(110101)_2 = 32 + 16 + 4 + 1 = (53)_{10}$$

Introduction of Number Systems

❖ Arithmetic operations with numbers in base r

- When a base other than the familiar base 10 is used, one must be careful to use only the r -allowable digits. Examples of addition, subtraction, and multiplication of two binary numbers are as follows:

$$\begin{array}{r} \text{augend: } 101101 \\ \text{addend: } + 100111 \\ \hline \text{sum: } 1010100 \end{array}$$

$$\begin{array}{r} \text{minuend: } 101101 \\ \text{subtrahend: } - 100111 \\ \hline \text{difference: } 000110 \end{array}$$

$$\begin{array}{r} \text{multiplicand: } 1011 \\ \text{multiplier: } * 101 \\ \hline \begin{array}{l} \nearrow 1011 \\ \rightarrow 0000 \\ \searrow 1011 \end{array} \\ \hline \text{Product } 110111 \end{array}$$

Binary divisions of $110 \div 11$

$$\begin{array}{r} 10 \\ 11 \overline{) 110} \\ \underline{11} \\ 000 \end{array}$$

Number-base Conversions

❖ Number-base Conversions

1. Convert decimal 41 to binary

First, 41 is divided by 2 to give an integer quotient of 20 and a remainder of $1/2$. Then the quotient is again divided by 2 to give a new quotient and remainder. The process is continued until the integer quotient becomes 0. The coefficients of the desired binary number are obtained from the remainders as follows:

		Integer Quotient		Remainder	Coefficient
$41/2$	=	20	+	$1/2$	$a_0=1$
$20/2$	=	10	+	0	$a_1=0$
$10/2$	=	5	+	0	$a_2=0$
$5/2$	=	2	+	$1/2$	$a_3=1$
$2/2$	=	1	+	0	$a_4=0$
$1/2$	=	0	+	$1/2$	$a_5=1$

Therefore, the answer is $(41)_{10} = (a_5 a_4 a_3 a_2 a_1 a_0)_2 = (101001)_2$

Number-base Conversions

❖ Number-base Conversions

2. Convert decimal 153 to octal.

The required base r is 8. First, 153 is divided by 8 to give an integer quotient of 19 and a remainder of 1. Then 19 is divided by 8 to give an integer quotient of 2 and a remainder of 3. Finally, 2 is divided by 8 to give a quotient of 0 and a remainder of 2. This process can be conveniently manipulated as follows:

		Integer Quotient		Remainder	Coefficient
153/8	=	19 (19*8=152)	+	1	$a_0=1$
19/8	=	2 (2*8=16)	+	3	$a_1=3$
2/8	=	0	+	2	$a_2=2$

Therefore, the answer is $(153)_{10} = (a_2 a_1 a_0)_2 = (231)_8$

Number-base Conversions

❖ Number-base Conversions

3. Convert $(0.6875)_{10}$ to binary.

First, 0.6875 is multiplied by 2 to give an integer and a fraction. Then the new fraction is multiplied by 2 to give a new integer and a new fraction. The process is continued until the fraction becomes 0 or until the number of digits has sufficient accuracy. The coefficients of the binary number are obtained from the integers as follows:

		Integer		Fraction	Coefficient
0.6875×2	=	1 (1.375)	+	0.375	$a_{-1}=1$
0.375×2	=	0 (0.75)	+	0.75	$a_{-2}=0$
0.75	=	1 (1.5)	+	0.5	$a_{-3}=1$
0.5	=	1	+	0	$a_{-4}=1$

Therefore, the answer is $(0.6875)_{10} = (0. a_{-1} a_{-2} a_{-3} a_{-4})_2 = (0.1011)_2$

Number-base Conversions

❖ Number-base Conversions

4. Convert $(0.513)_{10}$ to octal.

		Integer		Fraction	Coefficient
0.513×8	=	4 (4.104)	+	0.104	$a_{-1}=4$
0.104×8	=	0 (0.832)	+	0.832	$a_{-2}=0$
0.832×8	=	6 (6.656)	+	0.656	$a_{-3}=6$
0.656×8	=	5 (5.248)	+	0.248	$a_{-4}=5$
0.248×8	=	1 (1.984)	+	0.984	$a_{-5}=1$
0.984×8	=	7 (7.872)	+	0.872	$a_{-6}=7$
0.872×8	=	6 (6.976)	+	0.976	$a_{-7}=6$

Therefore, the answer is $(0.513)_{10} = (0. a_{-1} a_{-2} a_{-3} a_{-4} \dots)_2 = (0.4065176 \dots)_8$

Number-base Conversions

❖ Number-base Conversions

Octal And Hexadecimal Numbers

- The conversion from and to binary, octal, and hexadecimal plays an important role in digital computers, because shorter patterns of hex characters are easier to recognize than long patterns of 1's and 0's. Since $2^3 = 8$ and $2^4 = 16$, each octal digit corresponds to three binary digits and each hexadecimal digit corresponds to four binary digits.
- **Octal to Binary:** The conversion from binary to octal is easily accomplished by partitioning the binary number into groups of three digits each, starting from the binary point and proceeding to the left and to the right.

$$(10\ 110\ 001\ 101\ 011\ .\ 111\ 100\ 000\ 110)_2 = (26153.7406)_8$$

2	6	1	5	3	7	4	0	6
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Number-base Conversions

❖ Number-base Conversions

- **Hexadecimal to Binary:** Conversion from binary to hexadecimal is similar, except that the binary number is divided into groups of four digits:

$$(10\ 1100\ 0110\ 1011\ .\ 1111\ 0000\ 0110)_2 = (2C6B.F02)_{16}$$

2	C	6	B	F	0	2
---	---	---	---	---	---	---

- Conversion from octal or hexadecimal to binary is done by reversing the preceding procedure. Each octal digit is converted to its three-digit binary equivalent. Similarly, each hexadecimal digit is converted to its four-digit binary equivalent.

Octal or hexadecimal numbers Importance

- ❖ **Most computer manuals use either octal or hexadecimal numbers to specify binary quantities.**
- Binary numbers are difficult to work with because they require three or four times as many digits as their decimal equivalents. For example, the binary number 111111111111 is equivalent to decimal 4095.
- Thus, the binary number 111111111111 has 12 digits and is expressed in octal as 7777 (4 digits) or in hexadecimal as FFF (3 digits). During communication between people (about binary numbers in the computer), the octal or hexadecimal representation is more desirable because it can be expressed more compactly with a third or a quarter of the number of digits required for the equivalent binary number.

Complements of Numbers

- ❖ **Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.**
- Simplifying operations leads to simpler, less expensive circuits to implement the operations.
- There are two types of complements for each base- r system:
 - The radix complement: The first is referred to as the r 's complement (2 's complement and 10 's complement).
 - The diminished radix complement: the $(r-1)$'s complement (1 's complement and 9 's complement).

Comparison between 1's and 2's complements

1's Complement	2's Complement
1. It can be easily obtained using an Inverter	1. It has to be arrived at by first obtaining the 1's complement and then adding one (1) to it
2. It requires two operations	3. Only one arithmetic operation is required
3. It is often used in logical manipulations for inversion operation	3. It is used only for arithmetic applications

Subtraction with Complements

❖ Subtraction with 1's Complements

- **Step 01:** Find 1's Complement of number to be subtracted
- **Step 02:** Perform the addition
- **Step 03:** If final carry is 1 then add it to the result obtained at step 2,
If final carry is 0 result obtained at step 2 is negative and in it the 1's complement form

❖ Subtraction with 2's Complements

- **Step01:** Find 2's Complement of number to be subtracted
- **Step 02:** Perform the addition
- **Step 03:** If final carry is generated then the result is positive and in its true form,
If final carry is not produced then the result is negative and in its 2's complement form

Subtraction with Complements

❖ Subtraction with 1's Complements

- Given the two binary numbers $X = 110101$ and $Y = 100101$; perform the subtraction $X - Y$ and by using 1's complements.

- 1's complement of $Y=100101$ is $Y'=011010$.

- Hence, $X = 110101$

- $Y' = 011010$

$$\begin{array}{r} 110101 \\ + 011010 \\ \hline 1001111 \\ \swarrow 1 \\ \hline 010000 \end{array}$$

The required difference is **10000**

- If there is no end carry, the answer is:

$$Y - X = - (1's \text{ complement of } X' + Y)$$

Subtraction with Complements

❖ Subtraction with Complements

- Given the two binary numbers $X = 1010100$ and $Y = 1000011$, perform the subtraction (a) $X - Y$ and (b) $Y - X$ by using 2's complements.

a) Here, $X = 1010100$ and $Y = 1000011$ and base $r = 2$

1. 2's complement of Y : $(1\text{'s complement of } Y)_2 + 1 = 0111100 + 1 = 0111101$

2. Sum X and 2's complement of Y : $1010100 + 0111101 = 10010001$

So, $X - Y = 0010001$

(b) Here, $X = 1010100$ and $Y = 1000011$ and base $r = 2$

1. 2's complement of X : $(1\text{'s complement of } X)_2 + 1 = 0101011 + 1 = 0101100$

2. Sum Y and 2's complement of X : $1000011 + 0101100 = 1101111$

3. There is no end carry, so $Y - X = - (2\text{'s complement of } 1101111)$

So, $Y - X = - 0010001$

Tutorial Solve

Problem 01: Perform subtraction on the following unsigned binary numbers using the 2's-complement of the subtrahend. Where the result should be negative, 2's-complement it and affix a minus sign.

(a) 11011-11001 (b) 110100 -10101 (c) 1011-110000 (d) 101010-101011

Solⁿ of (a)

$$X = 11011 \quad Y = 11001$$

$$X = \quad 11011$$

$$2's \text{ Complement of } Y = + \underline{00111}$$

$$Sum = \quad 100010$$

$$\text{Discard end carry } 2^5 = - \underline{100000}$$

$$\text{Answer : } X - Y = \quad 00010$$

Solⁿ of (b)

$$X = 110100 \quad Y = 10101$$

$$X = \quad 110100$$

$$2's \text{ Complement of } Y = + \underline{101011}$$

$$Sum = \quad 1011111$$

$$\text{Discard end carry } 2^5 = - \underline{1000000}$$

$$\text{Answer : } X - Y = \quad 011111$$

Tutorial Solve

Problem 01: Perform subtraction on the following unsigned binary numbers using the 2's-complement of the subtrahend. Where the result should be negative, 2's-complement it and affix a minus sign.

(a) 11011-11001 (b) 110100 -10101 (c) 1011-110000 (d) 101010-101011

Solⁿ of (c)

$$X = 1011 \quad Y = 110000$$

$$X = \quad 1011$$

$$2's \text{ Complement of } Y = + \underline{010000}$$

$$Sum = \quad 011011$$

There is no end carry

$$\text{Answer : } Y - X = - \quad 100101$$

Solⁿ of (d)

$$X = 101010 \quad Y = 101011$$

$$X = \quad 101010$$

$$2's \text{ Complement of } Y = + \underline{010101}$$

$$Sum = \quad 111111$$

There is no end carry

$$\text{Answer : } Y - X = - \quad 000001$$

Complements of Numbers

- ❖ **Complements are used in digital computers to simplify the subtraction operation and for logical manipulation.**
- Simplifying operations leads to simpler, less expensive circuits to implement the operations.
- There are two types of complements for each base- r system:
 - The radix complement: The first is referred to as the r 's complement (2 's complement and 10 's complement).
 - The diminished radix complement: the $(r-1)$'s complement (1 's complement and 9 's complement).

Complements of Numbers

❖ Radix Complement

- The r 's complement of an n -digit number N in base r is defined as:

$$\begin{aligned}\text{Radix Complement} &= (r^n - N) && \text{for } N \neq 0 \\ &= 0 && \text{for } N = 0\end{aligned}$$

- The 10's complement of 246700 = $(10^6 - 246700) = \mathbf{753300}$
- The 2's complement of 1101100 = $(2^7 - 1101100) = 10000000 - 1101100 = \mathbf{0010100}$
- The complement of the complement restores the number to its original value**
 r 's complement of N is: $(r^n - N)$
If, $N' = (r^n - N)$, then r 's complement of N' is: $(r^n - (r^n - N)) = N$

Complements of Numbers

❖ Diminished Radix Complement

- Given a number N in base r having n digits, the $(r - 1)$'s complement of N :

$$\text{Diminished Radix Complement} = (r^n - 1) - N$$

- 9's complement of N is: $(10^n - 1) - N$
- The 9's complement of 546700 = $(10^6 - 1) - 546700 = 453299$
- For binary numbers, $r = 2$ and $(r - 1) = 1$, so the 1's complement of N is:
- 1's complement of N is: $(2^n - 1) - N$
- The 1's complement of 1011000 = $(2^7 - 1)_2 - 1011000 = (10000000 - 1) - 1011000$
= $1111111 - 1011000 = 01001110$
- The 1's complement of a binary number is formed by changing 1's to 0's and 0's to 1's.**
- The $(r - 1)$'s complement of octal or hexadecimal numbers is obtained by subtracting each digit from 7 or F (decimal 15), respectively.**

Subtraction with Complements

❖ Subtraction with Complements

The subtraction of two n -digit unsigned numbers $M - N$ in base r can be done as follows:

1. Add the minuend M to the r 's complement of the subtrahend N .

$$M + (r^n - N) = (M - N) + r^n$$

2. If $M \geq N$, the sum will produce an end carry r^n , which can be discarded; what is left is the result $M - N$.
3. If $M < N$, the sum does not produce an end carry and is equal to $r^n - (N - M)$, which is the r 's complement of $(N - M)$. To obtain the answer in a familiar form, take the r 's complement of the sum and place a negative sign in front.

Complements of Numbers

❖ Subtraction with Complements

- Using 10's complement, subtract : $72532 - 3250$.

Here, $M = 72532$; and $N = 03250$ and base, $r = 10$

1. 10's complement of N : $(r^n - N) = 10^5 - 3250 = 96750$
2. Sum M and 10's complement of N : $72532 + 96750 = 169282$
3. $M \geq N$, Discard end carry (r^n): $169282 - 100000 = 69282$

- Using 10's complement, subtract : $3250 - 72532$.

Here, $M = 03250$; and $N = 72532$ and base, $r = 10$

1. 10's complement of N : $(r^n - N) = 10^5 - 72532 = 27468$
2. Sum M and 10's complement of N : $3250 + 27468 = 30718$
3. $M < N$, there is no end carry, the answer is: - (10's complement of 30718)
10's complement of 30718: $10^5 - 30718 = -69282$

Tutorial Solve

Problem 05: Why are Digital Systems important?

Solⁿ

- It is well suited for numerical and non-numerical information processing.
- Information processing can use a general-purpose system (computer).
- The finite number of values in a digital signal is represented by a vector of signals with just 2 values (binary signals).
- Digital signals are quite insensitive to variations of component variable values.
- Numerical digital systems can be made more accurate by increasing the number of digits used in the representation.
- Complex digital systems are built as integrated circuits composed of a large number of very simple devices.
- It is possible to select among different implementations of systems that trade off speed and amount of hardware.

Tutorial Exercise

Problem 01: Convert the following numbers with the indicated bases to decimal:

(a) $(4310)_5$ (b) $(198)_{12}$

(c) $(435)_8$ (d) $(345)_6$

Problem 02: What is the largest binary number that can be expressed with 16 bits?
What are the equivalent decimal and hexadecimal numbers?

Problem 03: Obtain the 1's and 2's complements of the following binary numbers:

(a) 00010000 (b) 00000000

(c) 11011010 (d) 10101010

(e) 10000101 (f) 11111111.

Problem 04: Find the 9's and the 10's complement of the following decimal numbers:

(a) 25,478,036 (b) 63, 325, 600

(c) 25,000,000 (d) 00,000,000.

Tutorial Exercise

Problem 06: Find the 16's complement of C3DF.

Problem 07: Perform subtraction on the given unsigned numbers using the 10's complement of the subtrahend. Where the result should be negative, find its 10's complement and affix a minus sign. Verify your answers.

(a) $4,637 - 2,579$ (b) $125 - 1,800$

(c) $2,043 - 4,361$ (d) $1,631 - 745$

Problem 08: Perform subtraction on the given unsigned binary numbers using the 2's complement of the subtrahend. Where the result should be negative, find its 2's complement and affix a minus sign.

(a) $10011 - 10010$ (b) $100010 - 100110$

(c) $1001 - 110101$ (d) $101000 - 10101$